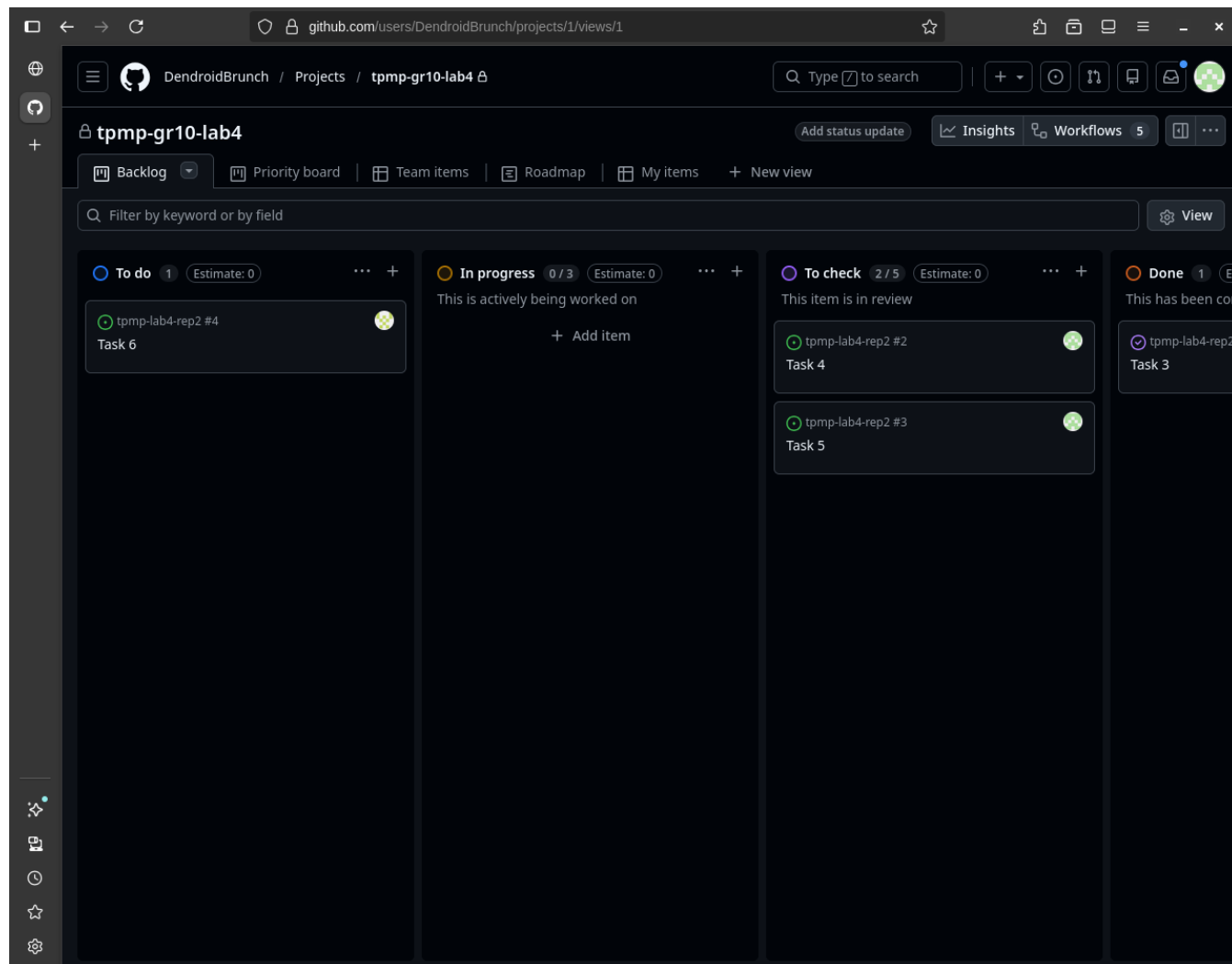


Лабораторная №4

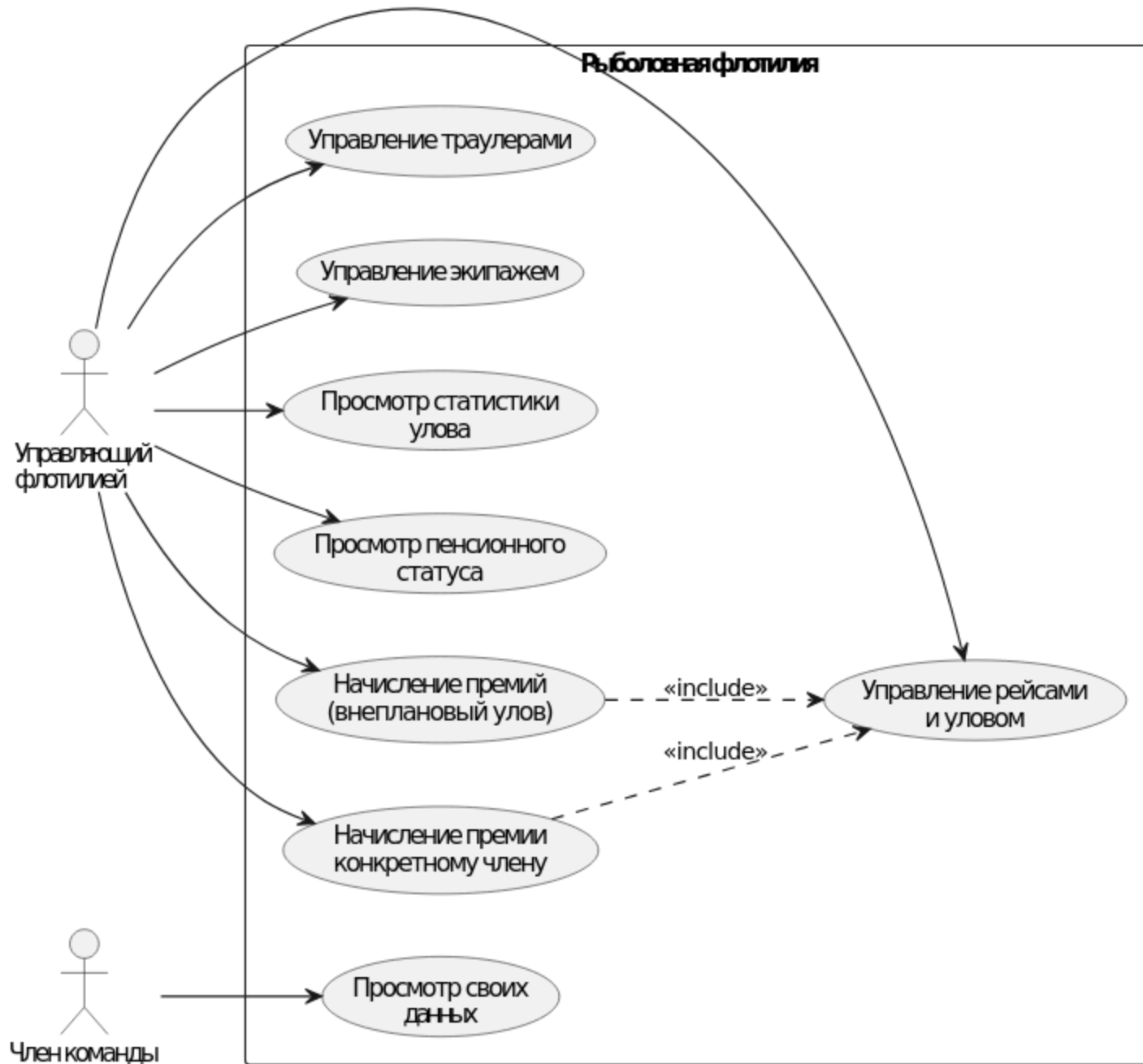
Размеркаванне задач у камандзе

- Жылко Барыс: 1, 2, 6
- Грынь Дзмітрый: 1, 3, 4, 5

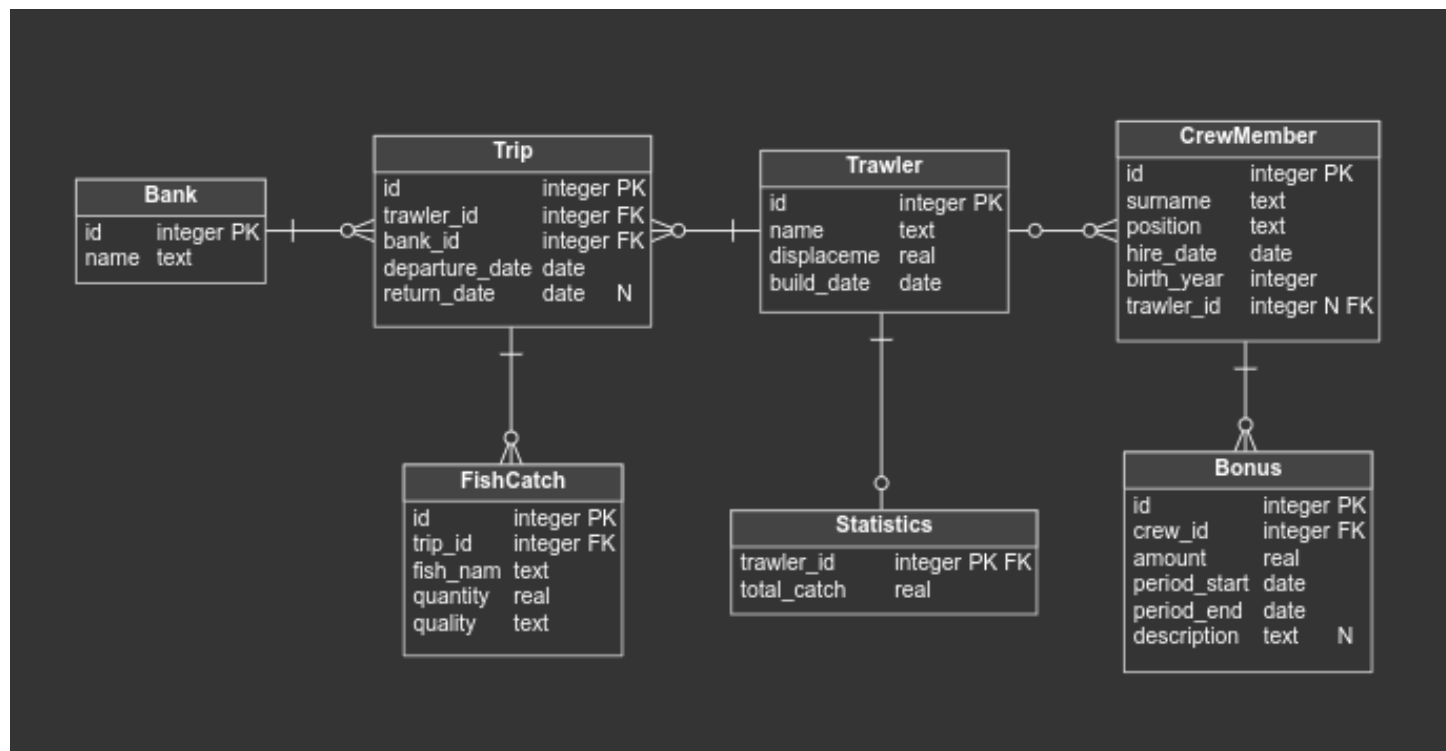


Патрабаванні

Дыяграма варыянтаў
выкарыстання

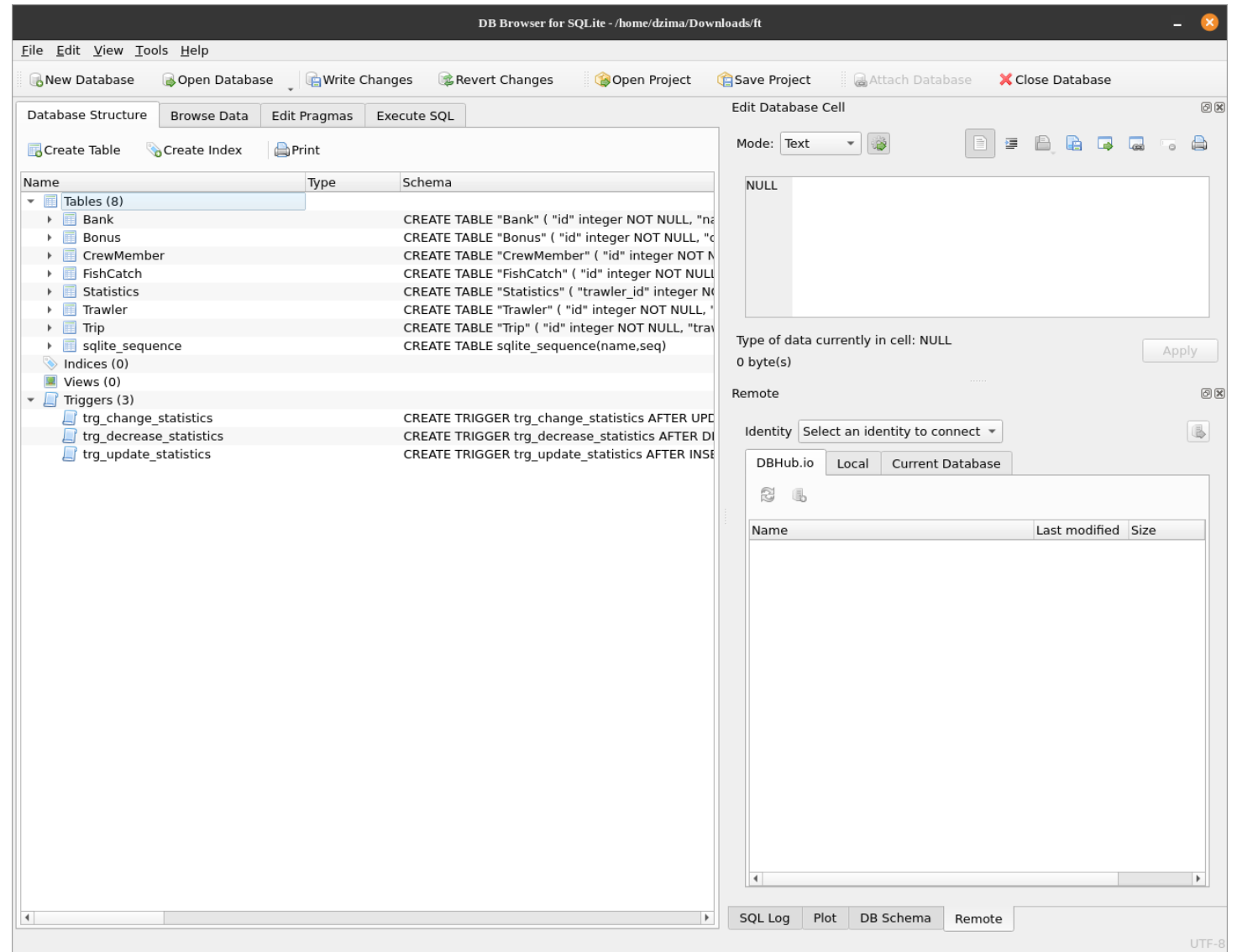


Структура БД (Vertabelo)



БД у SQLitebrowser

- Bank
- Bonus
- Trawler
- CrewMember
- FishCatch
- Statistics
- Trip



auth.h

- Аўтэнтыфікацыя
- Ролі

Code

Blame

43 lines (32 loc) · 913 Bytes



```
1  #ifndef AUTH_H
2  #define AUTH_H
3
4  #include <string>
5
6  enum class UserRole {
7      MANAGER,
8      CREW
9  };
10
11  struct User {
12      int id;
13      std::string username;
14      std::string password_hash;
15      UserRole role;
16      int crew_id;
17  };
18
19  class Auth {
20  private:
21      User current_user;
22      bool authenticated;
23
24      bool verifyPassword(const std::string& input, const std::string& hash);
25
26  public:
27      std::string hashPassword(const std::string& password);
28
29      Auth();
30
31      bool login(const std::string& username, const std::string& password);
32      void logout();
33      bool isAuthenticated() const { return authenticated; }
34      UserRole getRole() const { return current_user.role; }
35      int getCurrentUserId() const { return current_user.id; }
36      int getCurrentCrewId() const { return current_user.crew_id; }
37      std::string getCurrentUsername() const { return current_user.username; }
38  };
39
40  extern Auth* g_auth;
41
42
43  #endif
```

bonus.h

- Сістэма падліку бонусаў

Code Blame 42 lines (35 loc) · 1.27 KB

Raw Copy Download Edit View Source

```
1  #ifndef BONUS_H
2  #define BONUS_H
3
4  #include <string>
5  #include <vector>
6
7  struct Bonus {
8      int id;
9      int crew_id;
10     double amount;
11     std::string period_start;
12     std::string period_end;
13     std::string description;
14 };
15
16 class BonusCalculator {
17 public:
18     static int calculateBonuses(const std::string& start_date,
19                               const std::string& end_date,
20                               double plan,
21                               double price);
22
23     static bool calculateBonusForCrew(int crew_id,
24                                     const std::string& start_date,
25                                     const std::string& end_date,
26                                     double plan,
27                                     double price);
28
29     static std::vector<Bonus> getAllBonuses();
30     static void printBonuses();
31
32 private:
33     static double calculateCrewCatch(int crew_id,
34                                     const std::string& start_date,
35                                     const std::string& end_date);
36     static bool insertBonus(int crew_id, double amount,
37                            const std::string& start_date,
38                            const std::string& end_date,
39                            const std::string& description);
40 };
41
42 #endif
```

database.h

- Злучэнне
- Запыты да базы
- Абнаўленне базы

```
10  class Database {
11      private:
12          sqlite3* db;
13          std::string db_path;
14
15      public:
16          Database(const std::string& path);
17          ~Database();
18
19          bool open();
20          void close();
21          bool execute(const std::string& sql);
22          bool executeScript(const std::string& scriptPath);
23
24          bool query(const std::string& sql,
25                    int (*callback)(void*, int, char**, char**),
26                    void* data);
27
28  class Statement {
29      private:
30          sqlite3_stmt* stmt;
31      public:
32          Statement(sqlite3* db, const std::string& sql);
33          ~Statement();
34          bool step();
35          void bind(int pos, int value);
36          void bind(int pos, const std::string& value);
37          void bind(int pos, double value);
38          void bind(int pos, const char* value);
39          int getInt(int col);
40          double getDouble(int col);
41          std::string getString(int col);
42          bool isValid() const { return stmt != nullptr; }
43      };
44
45      std::unique_ptr<Statement> prepare(const std::string& sql);
46
47      bool beginTransaction();
48      bool commit();
49      bool rollback();
50
51      sqlite3* getHandle() { return db; }
52  };
53
54  extern Database* g_db;
55
56  #endif
```


statistics.h

- Статыстыка кожнага траўлера

```
1  #ifndef STATISTICS_H
2  #define STATISTICS_H
3
4  #include <sqlite3.h>
5  #include <string>
6
7  struct Statistics {
8      int trawler_id;
9      double total_catch;
10 };
11
12 ✓ class StatisticsService {
13     public:
14         static bool updateStatistics(int trawler_id);
15         static double getTotalCatch(int trawler_id);
16         static void printAllStatistics();
17
18     private:
19         static double calculateTotalCatch(int trawler_id);
20 };
21
22 void setupStatisticsTriggers(sqlite3* db);
23
24 #endif
```

trip.h

- Структуры

```
8  ✓ struct FishCatch {
9      std::string fish_name;
10     double quantity;
11     std::string quality;
12 };
13
14  ✓ struct Trip {
15     int id;
16     int trawler_id;
17     int bank_id;
18     std::string departure_date;
19     std::string return_date;
20     std::vector<FishCatch> catches;
21 };
22
23  ✓ struct Trawler {
24     int id;
25     std::string name;
26     double displacement;
27     std::string build_date;
28 };
29
30     struct Bank {
31         int id;
32         std::string name;
33     };
34
35  ✓ struct CrewMember {
36     int id;
37     std::string surname;
38     std::string position;
39     std::string hire_date;
40     int birth_year;
41     int trawler_id;
42
43     int getAge() const;
44     bool isRetirementDue(const std::string& date) const;
45 };

```

trip.h

- CRUD-операції для всіх елементів

```
47  class TripService {
48      public:
49          bool addTrip(const Trip& trip);
50          std::vector<Trip> getTripsByTrawler(int trawler_id,
51                                             const std::string& start_date,
52                                             const std::string& end_date);
53
54          void printTrawlerTrips(int trawler_id, const std::string& start, const std::string& end);
55          void printBankCatches(int bank_id);
56          void printMaxLowQualityBank();
57          void printTopTrawlerInfo();
58          void printCrewRetirementDue(const std::string& date);
59
60          bool insertTrawler(const Trawler& trawler);
61          bool updateTrawler(int id, const Trawler& trawler);
62          bool deleteTrawler(int id);
63
64          bool insertCrewMember(const CrewMember& member);
65          bool updateCrewMember(int id, const CrewMember& member);
66          bool deleteCrewMember(int id);
67
68          bool insertBank(const Bank& bank);
69          bool updateBank(int id, const Bank& bank);
70          bool deleteBank(int id);
71
72          std::vector<Trawler> getAllTrawlers();
73          std::vector<CrewMember> getAllCrewMembers();
74          std::vector<Bank> getAllBanks();
75          CrewMember getCrewMemberById(int id);
76          Trawler getTrawlerById(int id);
77
78          void printTrawlersTable();
79          void printCrewTable();
80          void printBanksTable();
81      };
82
83  #endif
```

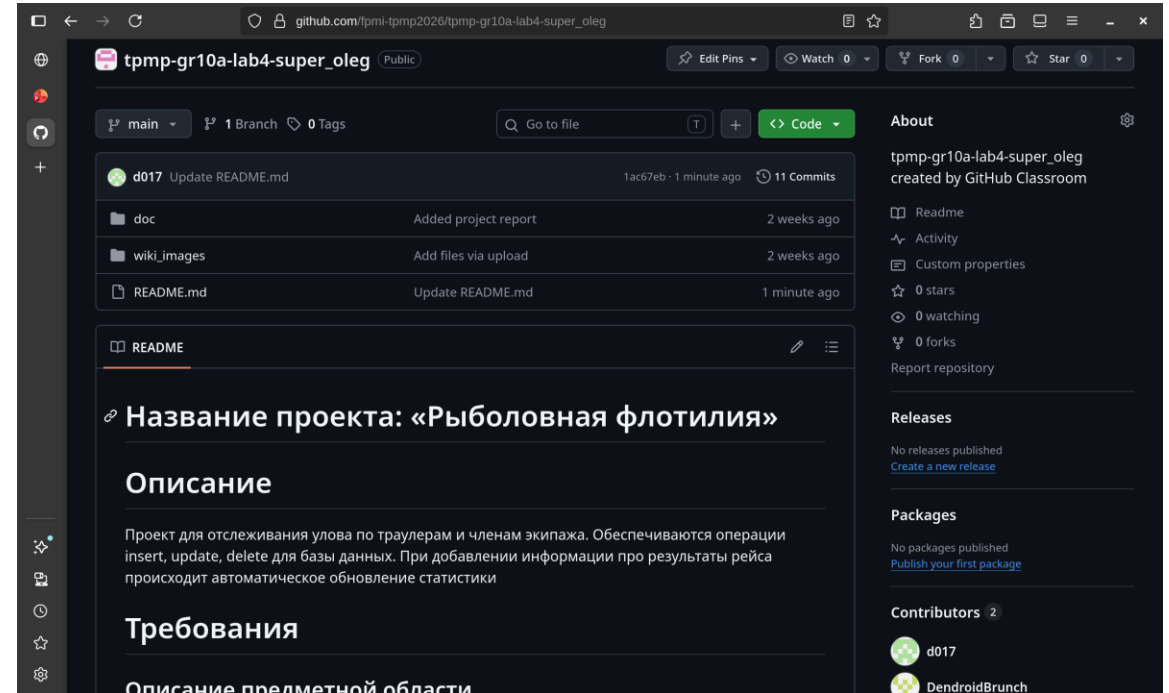
utils.h

- Дапаможныя функцыі

```
1  #ifndef UTILS_H
2  #define UTILS_H
3
4  #include <string>
5  #include <vector>
6  #include <ctime>
7  #include <iomanip>
8  #include <sstream>
9
10 namespace Utils {
11     std::string getCurrentDate();
12     bool isValidDate(const std::string& date);
13     int daysBetween(const std::string& d1, const std::string& d2);
14
15     std::string inputString(const std::string& prompt);
16     int inputInt(const std::string& prompt, int min = -1, int max = -1);
17     double inputDouble(const std::string& prompt, double min = -1, double max = -1);
18     char inputChar(const std::string& prompt);
19
20     void printHeader(const std::string& title);
21     void printSeparator(int len = 80);
22     void clearScreen();
23     void waitForEnter();
24     void printTableRow(const std::vector<std::string>& cols, int widths[]);
25 }
26
27 #endif
```

Рэпазіторый 1

- Github wiki
- Пастаноўка задачы
- Патрабаванні
- Дыяграмы



Рэпазіторый 2

- Код праекта
- Тэсты
- Github actions

